Statistics for Computer Scientists

Lecture 5: Sampling distributions

10 October 2025

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Where are we

Lecture	Topic	Book
5 Sept	Introduction, statistics fundamentals	Chapter 1
12 Sept	Descriptive statistics	Chapter 2
19 Sept	Bivariate relationships	Chapter 3
26 Sept	Data collection	Chapter 4
10 Oct	Sampling distributions	Chapter 7
17 Oct	Estimation	Chapter 8
31 Oct	Significance testing	Chapter 9
7 Nov	Comparing two groups	Chapter 10
14 Nov	Categorical data analysis	Chapter 11
21 Nov	Linear regression	Chapter 12
28 Nov	Multiple regression	Chapter 13
5 Dec	Tba	
12 Dec	Tba	

Today's lecture

- Recap lecture 4

- Sampling distributions
  - Proportion
  - Mean
  - Relevance
- Probability tables
- Bootstrapping

Recap

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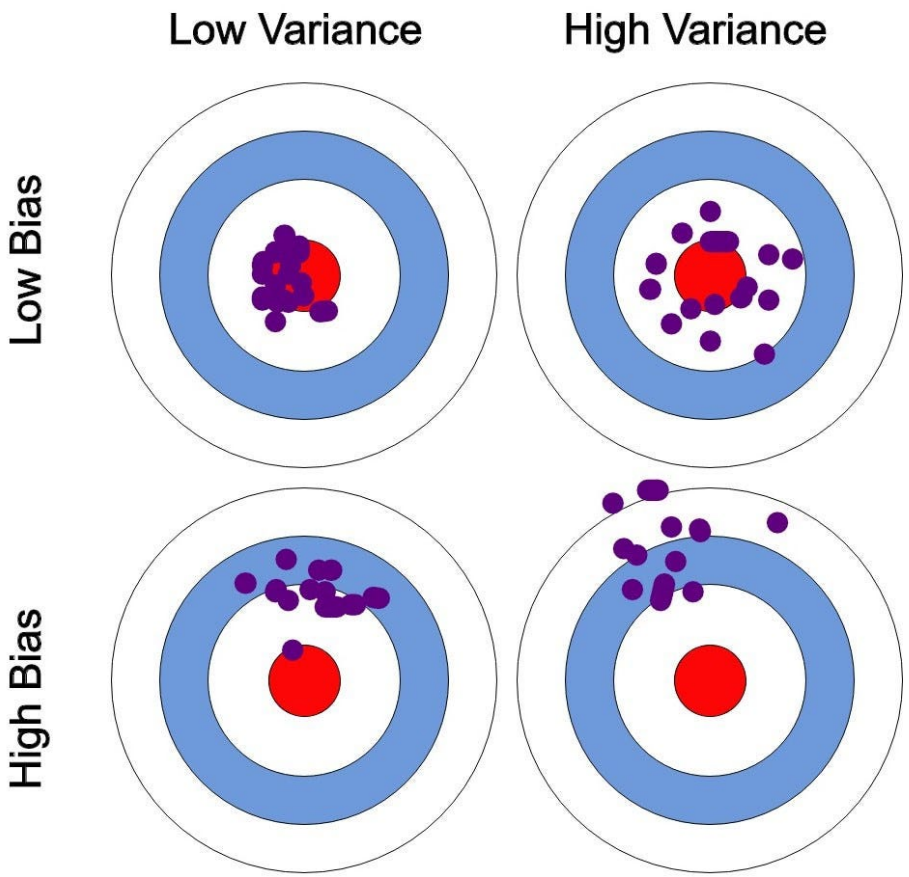
Recap

- Data collection also part of statistics
- Probability sampling
  - Simple random, systematic, stratified, cluster
  - Multistage sampling

- Bias
  - Systematic error in statistical result
  - Sampling bias, nonresponse bias, response bias



Sampling distribution: proportion

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Proportion

- Election with two voting options: Whitman vs Brown
  - Categorical nominal (binary) variable

- **Parameter of interest**: population proportion of people voting for Brown
- **Statistic**: sample proportion of people voting for Brown

- What does population distribution look like?
- What does sample/data distribution look like?

Sampling distribution

- Sampling distribution of a **statistic** describes all potential values that could be observed for the statistic when taking a random sample of size n from the population

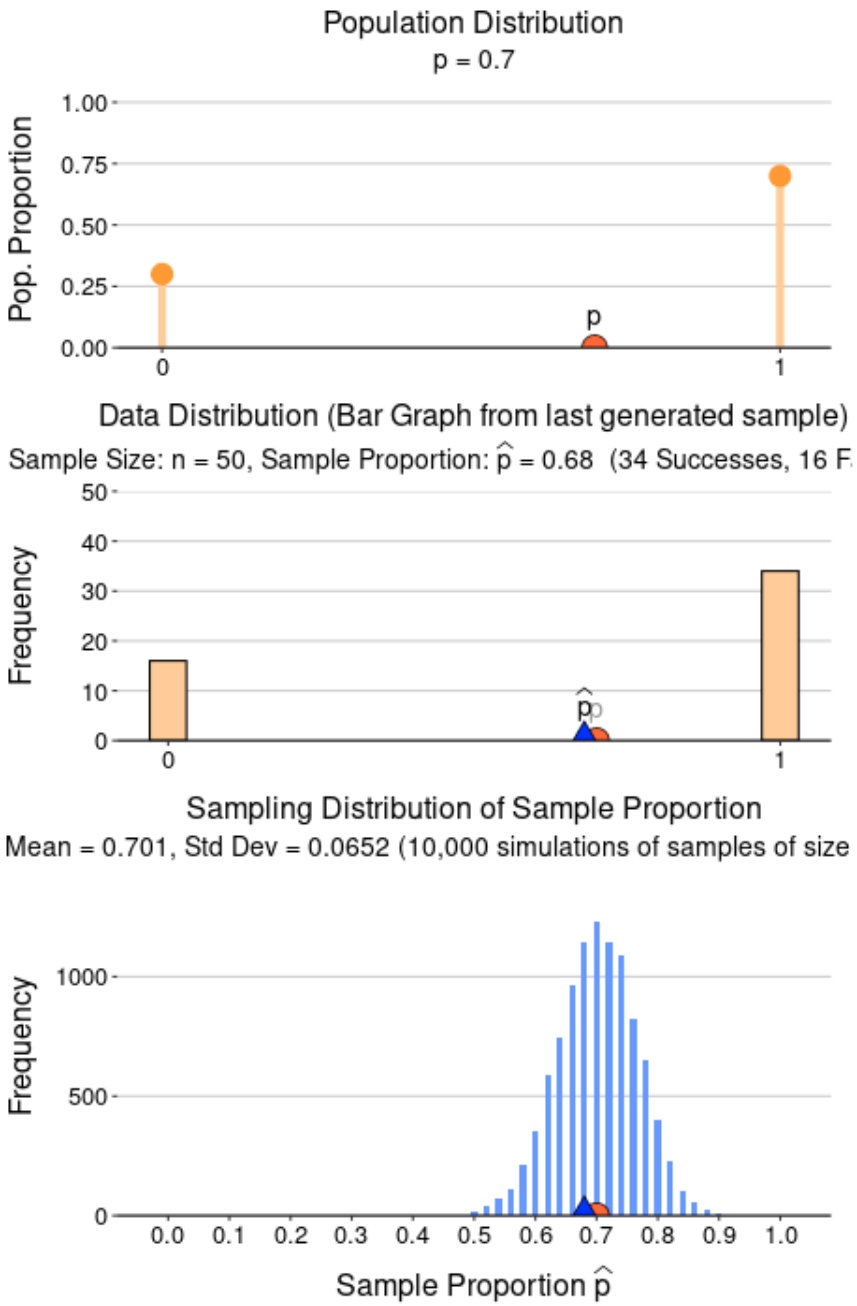
- Statistic is the sample proportion
- What kind of variable? What kind of distribution?
  - A statistic is also a random variable!

- https://istats.shinyapps.io/SampDist_Prop/

Sampling distribution

- Statistic (mean or proportion) is a continuous variable, no matter the variable type of the original variable

- No matter the shape of the population distribution, the **sampling distribution** approaches a normal distribution!
  - When sample size is large enough
  - Assuming random sampling



Describing sampling distr.

- Variable is categorical binary, described by a single statistic in the population: $\pi = 0.7$
  - But also has standard deviation! $\sigma = \sqrt{\pi(1-\pi)} = 0.46$

- $\hat{\pi}$ follow a normal distribution, with mean and standard deviation
  - $\mu_{\hat{\pi}} = \pi = 0.7$
  - $\sigma_{\hat{\pi}} = \frac{\sigma}{\sqrt{n}} = \sqrt{\frac{\pi(1-\pi)}{n}} = \frac{0.46}{\sqrt{50}} = 0.065$

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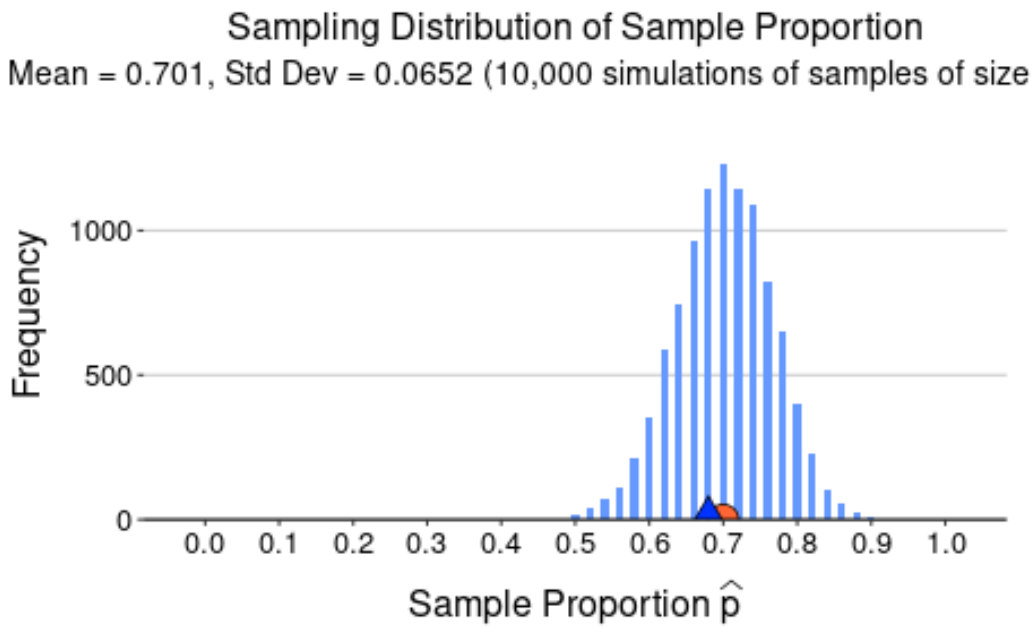
Toward inference…

Given a sampling distribution with $\mu_{\hat{\pi}} = 0.7$ and $\sigma_{\hat{\pi}} = 0.065$

What is the probability of finding $\hat{\pi} = 0.8$ or more extreme?

After finding sample statistic, can calculate probability of it coming from specific population! Significance testing



Standard error (important, not in book)

- Book: "standard deviation of the sampling distribution"
- Usually called **standard error**, also in software

- Population standard deviation: σ (usually unknown)
- Sample standard deviation: s (calculated from sample)
- Standard error: $\sigma_{\bar{x}}$ or $\sigma_{\hat{\pi}} = \frac{\sigma}{\sqrt{n}}$
  - But since σ usually unknown, we estimate standard error by: $se = \frac{s}{\sqrt{n}}$

Sampling distribution: mean

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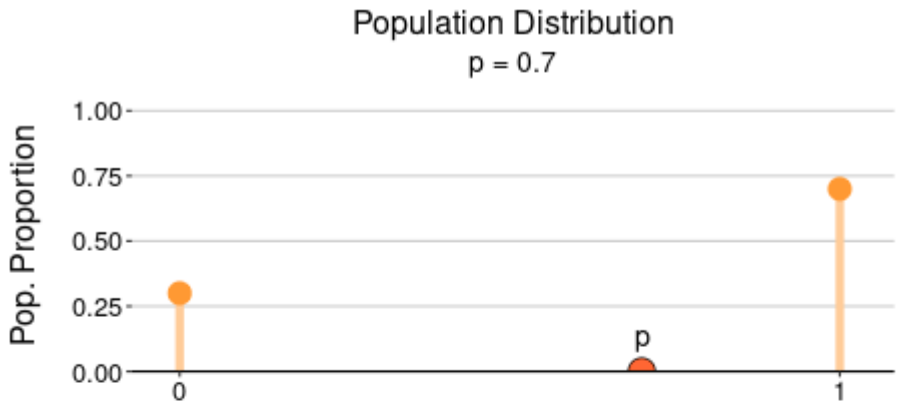
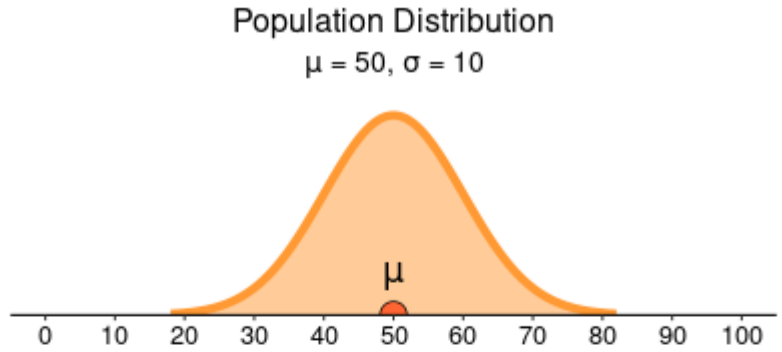
Mean vs proportion

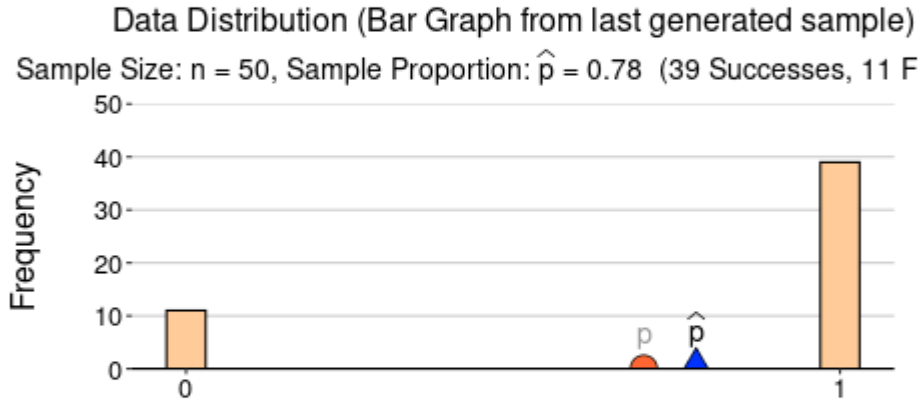
- Categorical variable/proportion: single parameter π, σ can be calculated from π
- Quantitative variable/mean: two parameters, μ and σ !

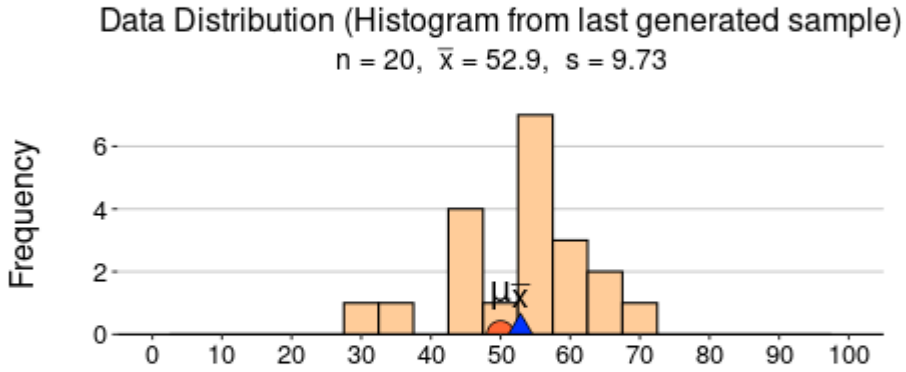
- Population and sample distributions look very different
  - Bar plot vs histogram

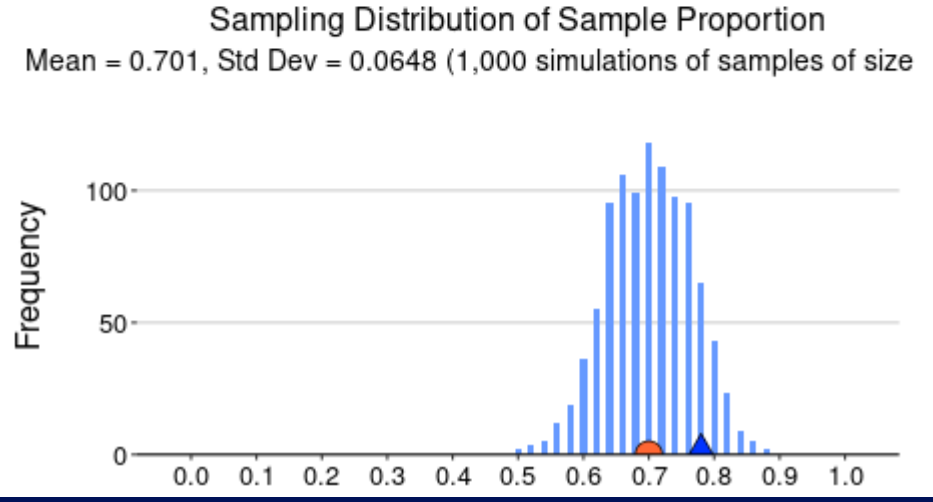
- Sampling distributions are always continuous!

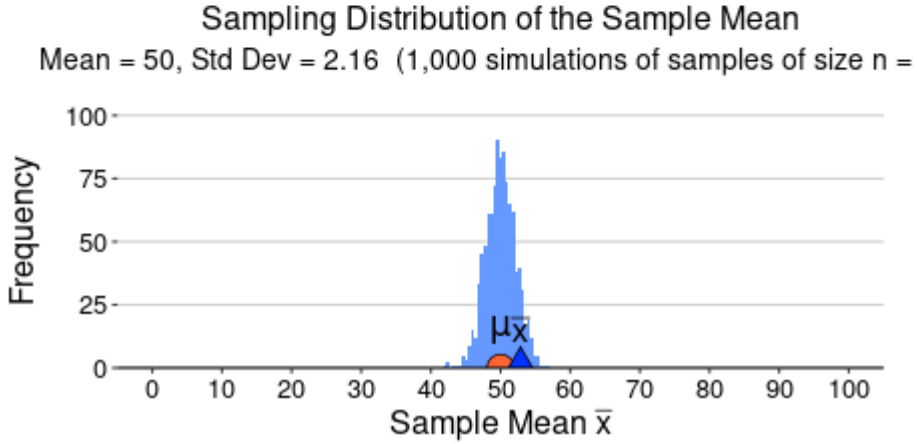

Population Distribution
p = 0.7
Pop. Proportion
1.00
0.75
0.50
0.25
0.00
p
0
1


Population Distribution
μ = 50, σ = 10
μ
0 10 20 30 40 50 60 70 80 90 100


Data Distribution (Bar Graph from last generated sample)
Sample Size: n = 50, Sample Proportion: p̂ = 0.78 (39 Successes, 11 F
Frequency
50
40
30
20
10
0
p
p̂
0
1


Data Distribution (Histogram from last generated sample)
n = 20, x̄ = 52.9, s = 9.73
Frequency
6
4
2
0
μx̄
0 10 20 30 40 50 60 70 80 90 100


Sampling Distribution of Sample Proportion
Mean = 0.701, Std Dev = 0.0648 (1,000 simulations of samples of size
Frequency
100
50
0
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0


Sampling Distribution of the Sample Mean
Mean = 50, Std Dev = 2.16 (1,000 simulations of samples of size n =
Frequency
100
75
50
25
0
μx̄
0 10 20 30 40 50 60 70 80 90 100
Sample Mean x̄

Describing sampling distr. of mean

- $\bar{x}$ (sample means) follow a normal distribution, described by mean and standard error
- $\mu_{\bar{x}} = \mu$ (on average, sample mean centered around population mean)
- $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ (standard error is std dev divided by square root of sample size)
- Since σ is usually unknown, estimate standard error by:
- $se = \frac{s}{\sqrt{n}}$

Central limit theorem

- Also covered in probability theory

- For a random sample of size n from a population having mean μ and standard deviation σ, as the sample size n increases, the sampling distribution of the sample mean $\bar{x}$ approaches a normal distribution

- (How do proportions play into this?)

Sample size

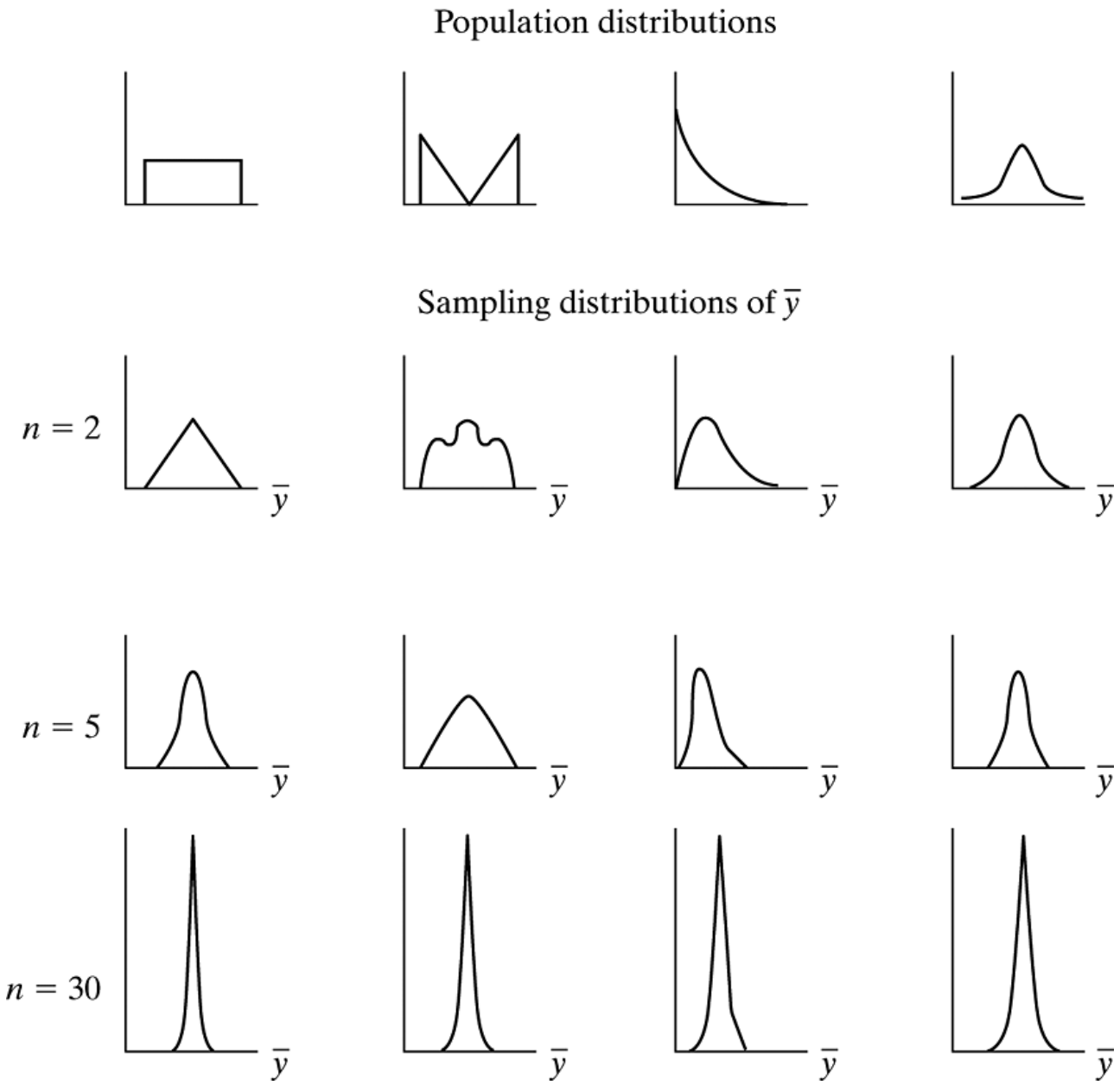
- For a random sample of size n from a population having mean μ and standard deviation σ, **as the sample size n increases**, the sampling distribution of the sample mean $\bar{x}$ approaches a normal distribution

- https://istats.shinyapps.io/sampdist_cont/
  - Real population data: delay of flights arriving at ATL
  - Shape of population distribution?
  - Which value for n to get a normal-ish sampling distribution? (zoom in for better view)

Rule of thumb for CLT: $n > 30$

- For a random sample of size n from a population having mean μ and standard deviation σ, **as the sample size n increases**, the sampling distribution of the sample mean $\bar{x}$ approaches a normal distribution

Population distribution	Sampling distribution for n < 30	Sampling distribution for n > 30
Normal	Normal	Normal
Skewed left	Skewed left	Normal
Uniform	Triangle-normal	Normal
Bernoulli	?	Normal


Population distributions
Sampling distributions of $\bar{y}$
$n = 2$
$n = 5$
$n = 30$
$\bar{y}$

Why do we care?

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Using the sampling distribution

- Usually interested in estimating population mean or proportion
- CLT states that the sampling distribution approaches the normal distribution (under assumptions)

- Calculate probability that your sample comes from a specific population
- P-value and hypothesis pairs
- “P-value is the probability of finding your sample statistic, IF the null hypothesis is true”
  - Null hypothesis states the hypothesized population

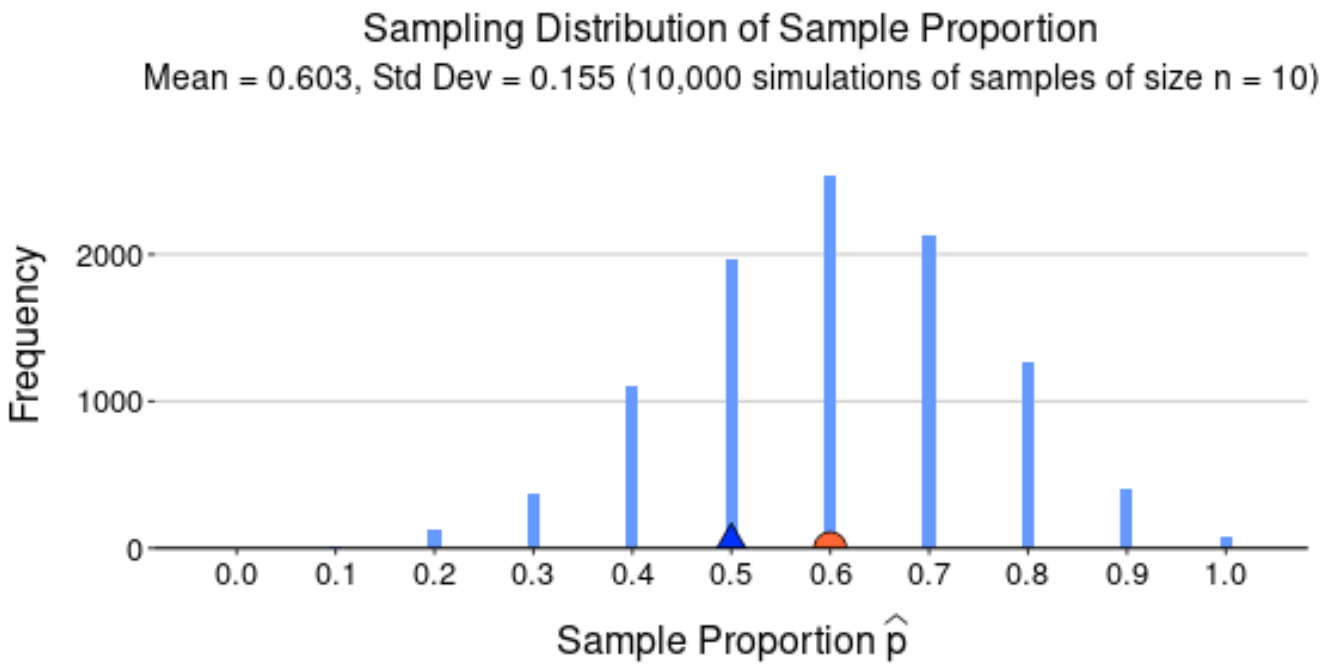
Using the sampling distribution

- Medicine vs placebo, which has higher survival rate?
  - Medicine group: 70% survival (n=10)
  - Placebo group: 60% survival (n=10)

- If medicine not better than placebo, both groups sample from same population
  - What is the probability of finding 70% survival in the sample, when the population survival is 60%?
  - Sample proportions normally distributed with $\mu_{\hat{\pi}} = 0.6$ and $se = \frac{\sigma}{\sqrt{n}} = \frac{\sqrt{0.6*0.4}}{\sqrt{10}} = 0.15$
  - Medicine group survival is $\frac{0.70-0.60}{0.15} = 0.67$ standard errors higher than mean
  - Can be turned into a probability!

Finding the probability

- https://istats.shinyapps.io/SampDist_Prop/



Continuous example

- Pizza company says their average delivery time is 20 minutes
- In my sample of $n = 30$ orders, I found $\bar{x} = 23$ minutes and s =11 minutes
- Is my sample significantly different from the claim? (Does my sample come from a population with $\mu = 20$?)

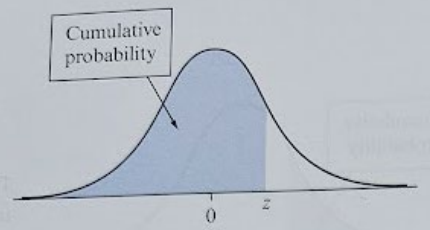
- Sampling distribution has $\mu_{\bar{x}} = 20$ and $se = \frac{11}{\sqrt{30}} = 2$
- Calculate z-score: $z = \frac{\bar{x}-\mu}{se} = \frac{23-20}{2} = 1.5$
  - Our sample mean is 1.5 standard errors higher than the mean

Continuous example continued

- We found $z = \frac{\bar{x}-\mu}{se} = \frac{23-20}{2} = 1.5$
- Book Table A: cumulative probabilities

- $P(z < 1.5) = 0.9332$
- $P(z > 1.5) = 1 - 0.9332 = 0.07$

- 7% probability of finding sample mean of 23 minutes **or higher** if the true mean is 20!



The cumulative probability for z is the area under the standard normal curve to the left of z

Table A Standard Normal Cumulative Probabilities (*continued*)

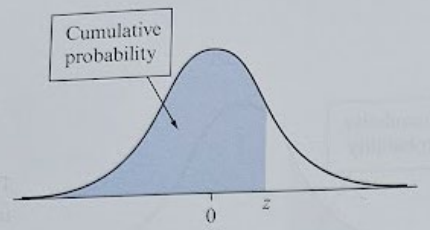
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706

Table A

- Shows left-tail probability for positive z
- To find right-tail probability, take complement

- Negative z-values: use symmetry!
- $P(z < -0.5) = P(z > 0.5) = 1 - P(z < 0.5)$

- Sketch the distribution when in doubt!
- Practice



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1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
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1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
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Bootstrapping

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CLT and statistics

- CLT holds for mean and proportion
- What about other statistics? (standard deviation, mean, correlation, etc)

- Assuming only sample data is available, not population data
- Solution: **bootstrap resampling**
  - Sample with replacement from the original sample, and of the same size as original sample

- https://istats.shinyapps.io/Boot1samp/

Bootstrap sampling distribution

- Sampling distribution: distribution of $\bar{x}$ when taking many samples from population
- Bootstrap sampling distribution: distribution of $\bar{x}$ when taking many bootstrap samples from **original sample**

- Bootstrap approximates the sampling distribution!
  - Limitations and quality of approximation?

Bootstrapping

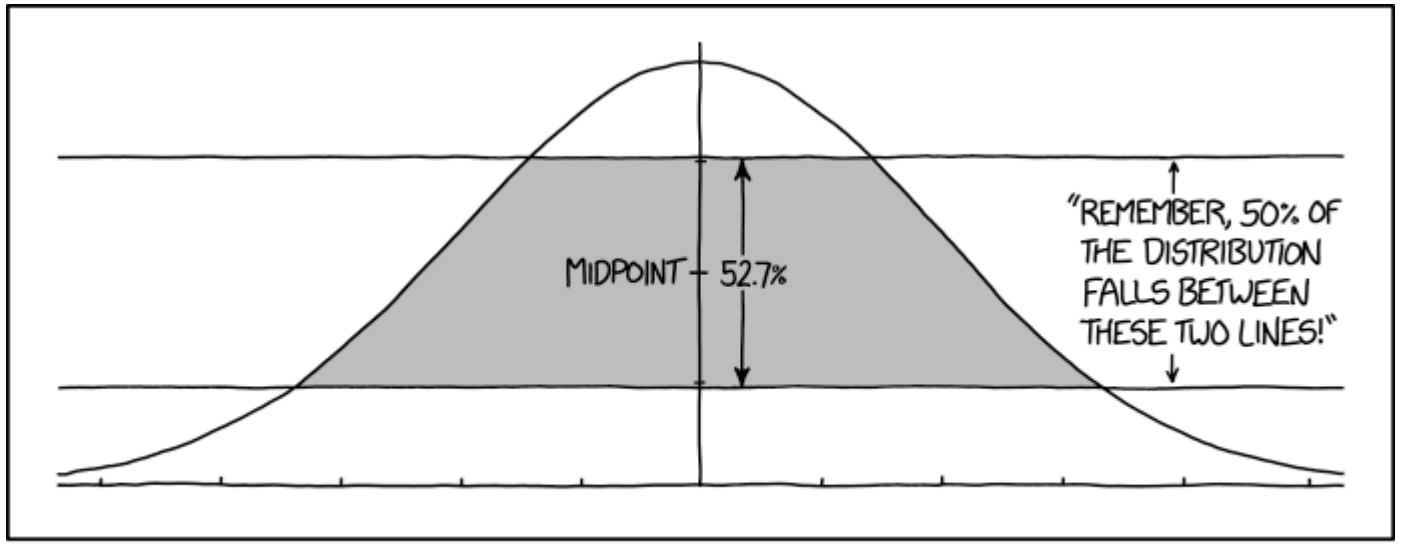
- Advantages
  - Simple
  - For any estimator and any distribution

- Disadvantages
  - Naïve use can give incorrect results
  - Sample must be representative
  - Assumptions

Wrapping up

- Read chapter 7 of the book (5+6 to catch up on probability)
- Tutorial: sampling distributions and bootstrapping

- Assignment grades next week
- Lecture 17 October: estimation
  - (chapter 8)



HOW TO ANNOY A STATISTICIAN
https://xkcd.com/2118/